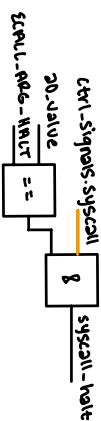
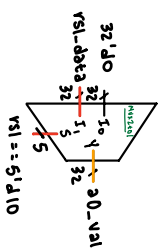
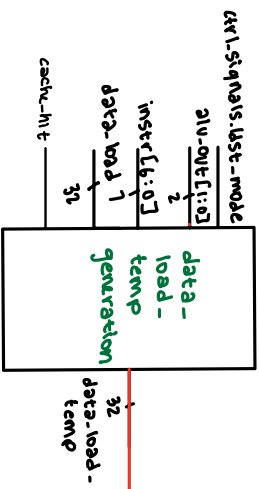


read from in
- registers

M5 M6 M7 M8

Writeback

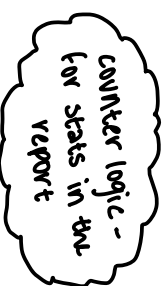
timeToWrite = ctrl_signals.rfWrite
rdToWrite = inster[11:7]



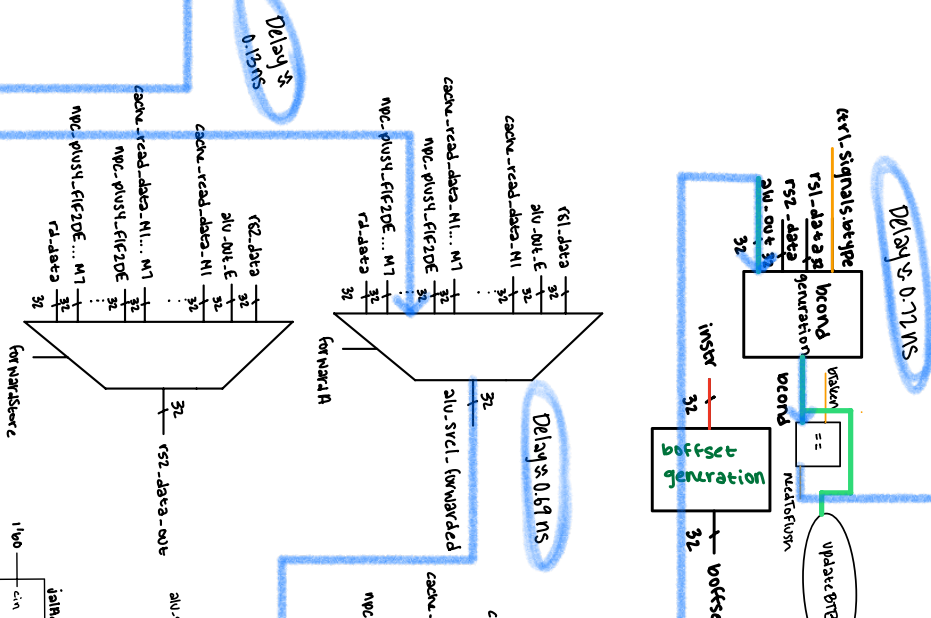
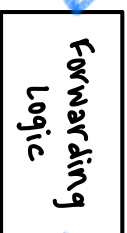
* timeToWrite, rdToWrite,
rd-data all get passed
back to the decode stage

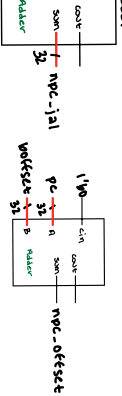
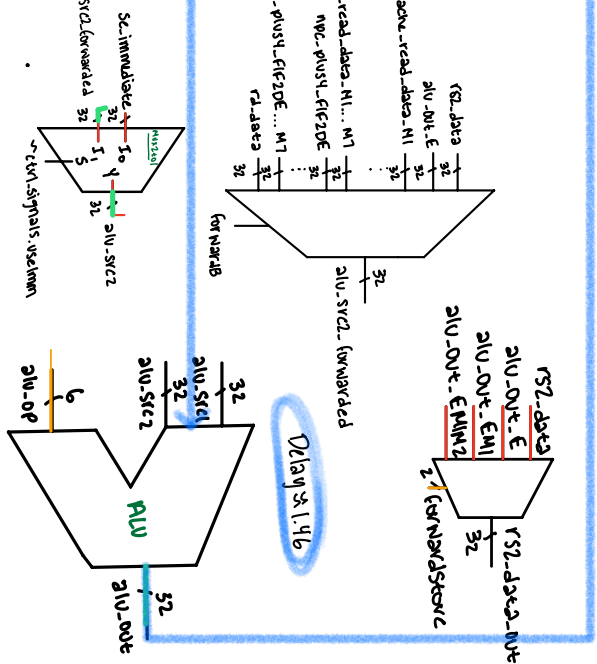
* Critical Path w/ Delays in BLUE

Effect

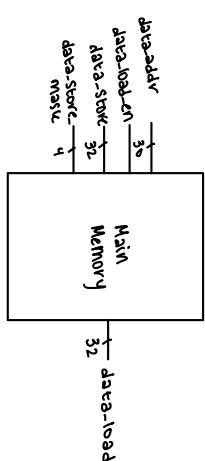
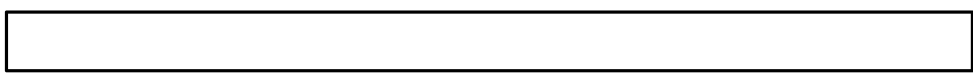


A diagram illustrating a transition or flow. It consists of two horizontal rectangular boxes, one above the other. A black arrow points downwards from the bottom center of the top box to the top center of the bottom box. The top box is white with a black border. The bottom box is light blue with a black border.

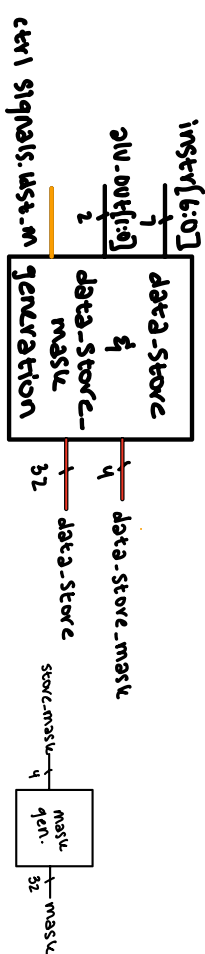




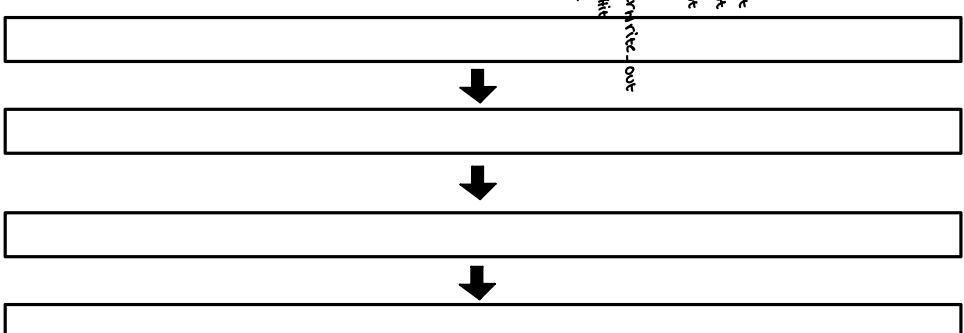
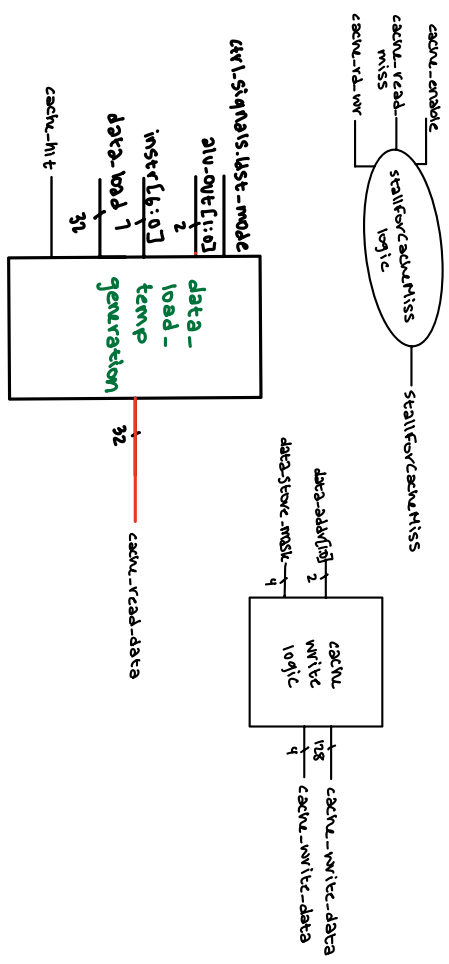
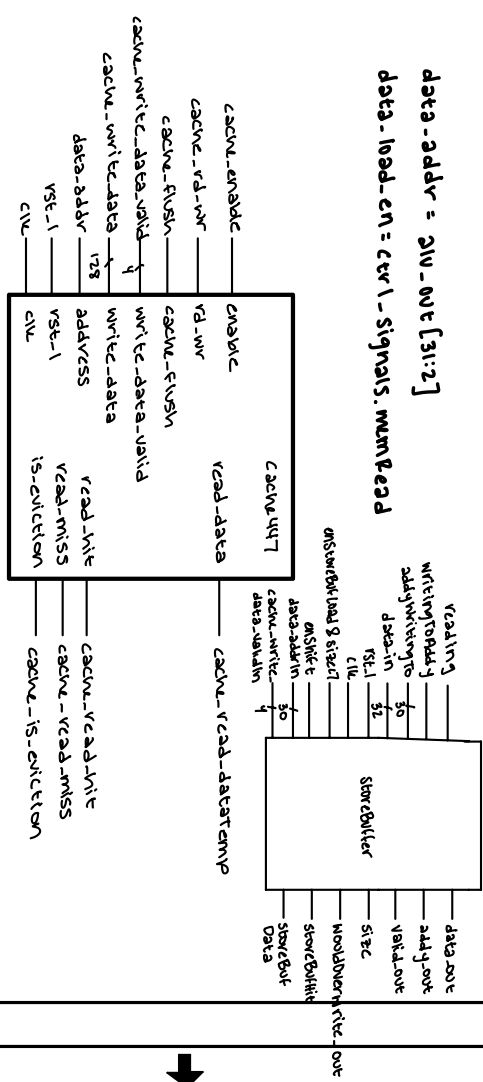
halting logic



- * signals that should be between memory stage
- 1) cache-hit
- 2) cache_read-data
- 3) alu-out
- 4) npc-plus4

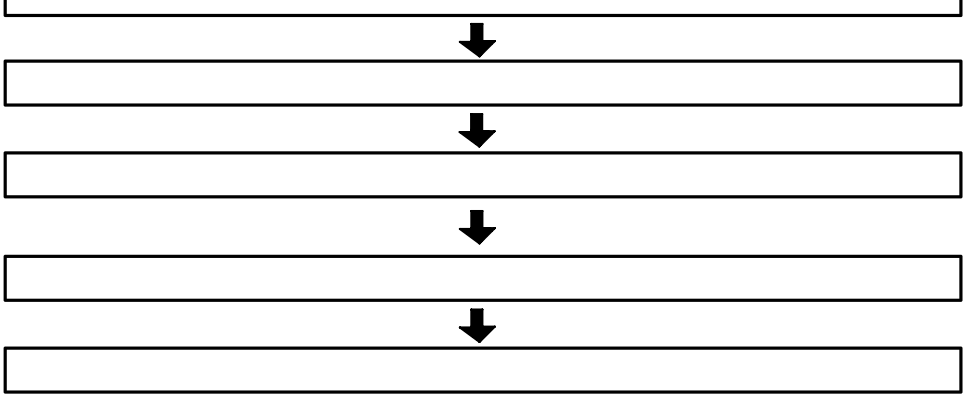


```
data_addr = in_out[31:2]
data_load_en = ctrl_signals.warmRead
```



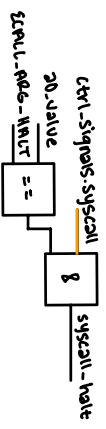
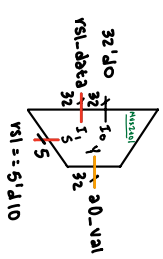
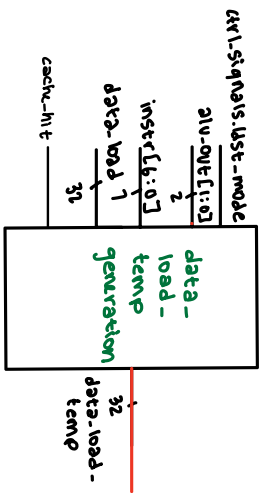
forwarded from in
e registers

M5 M6 M7 M8



Writeback

timeToWrite = ctrl_signals.rfWrite
rdToWrite = inster[11:7]



* timeToWrite, rdToWrite,
rd-data all get passed
back to the decode stage